

Reinforcement Learning-Based Agent for the Imperfect Information Game *Junqi*

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Abstract

This project develops a full reinforcement learning framework for the imperfect-information board game *Junqi* (Chinese Military Chess). A complete environment implementing the true game rules was created, followed by the design of two Deep Q-Learning (DQN) agents: (1) a Placement Agent for initial formation planning and (2) a Reasoning & Move Agent for gameplay under uncertainty. The system integrates belief-based inference, multi-stage decision making, and reward shaping that encourages information gain, deception, and strategic structure. Experiments show the emergence of robust formations and tactically meaningful behaviors that resemble human strategies.

1 Project Overview

Chinese Military Chess (*Junqi*) presents a challenging domain for reinforcement learning due to its large branching factor, heterogeneous movement rules, asymmetric piece interactions, and, most critically, its partially observable nature: piece identities remain hidden until battle or logically inferred. To address these challenges, this project develops a full-stack reinforcement learning system that spans environment modeling, multi-stage decision-making, belief-based inference, and tailored reward shaping. The system integrates three major components: (1) a high-fidelity game environment that encodes all rules of *Junqi*; (2) a two-stage Deep Q-Learning (DQN) architecture for formation planning and gameplay; and (3) an auxiliary inference module that maintains and updates probabilistic beliefs of hidden enemy ranks.

Environment and Rule Modeling. A complete Python-based simulation of *Junqi* was implemented from the ground up, faithfully modeling all structural and rule-level properties of the game. The environment supports a full 13×5 board with camp zones, railways and multi-step railway traversal, blockable intersections, and the asymmetric movement capabilities of different pieces. Combat resolution follows the official hierarchy (e.g., bombs causing mutual destruction, engineers clearing mines, and rank-based elimination). Crucially, the environment maintains hidden information by masking enemy ranks and only revealing identities upon engagement or logical inference. The action space is discretized via an indexing scheme that maps each (piece, destination) pair to a unique action ID, enabling the agent to operate on a fixed-size discrete space despite the variability and constraints of legal moves. This environment serves as the training ground for all agents and provides interfaces for state encoding, belief propagation, reward computation, and episodic self-play.

Two-Stage Deep Q-Learning Framework. Given the natural separation of *Junqi* into a hidden-information *placement stage* and a strategic *move-and-attack stage*, the learning problem is decomposed into two specialized DQN agents.

- **Placement Agent.** The placement phase is formulated as a sequential decision process where the agent places one piece at a time on its half of the board. The agent receives a state embedding that encodes the partially completed formation, remaining pieces, and positional constraints. A dueling DQN with noisy layers is trained to evaluate long-term formation robustness, using reward signals derived from formation validity, structural integrity (e.g., mine belts, commander fortification), and eventual game outcomes. This enables the agent to learn deceptive and strategically sound initial layouts rather than merely random or greedy placements.
- **Reasoning & Move Agent.** Once the game enters the move-and-attack phase, a second DQN agent controls piece movement and tactical decisions. Its state representation includes

visible board information, historical battle outcomes, estimated enemy ranks, and extracted spatial features. A belief-aware Q-network integrates both positional features and probabilistic rank distributions, enabling decisions that simultaneously improve positional advantage, avoid catastrophic trades, and actively probe uncertain enemy pieces.

This two-agent pipeline mirrors human play: strategic formation planning first, followed by tactical reasoning and combat execution.

Belief-Based Inference Module. A key component of the system is a dedicated inference environment and DoubleDQN-based inference model that predicts hidden enemy identities. The belief tensor maintains a probability distribution over all possible ranks for each unrevealed piece. Beliefs are updated using movement patterns, consistency rules, historical engagements, and domain-specific priors (e.g., mines being immobile, bombs not occupying choke points). The inference agent is rewarded for accurate predictions and penalized for implausible rank assignments, enabling it to learn nontrivial inference patterns that guide the move agent toward more informed and safer decisions.

Reward Shaping and Strategic Guidance. The project introduces a multi-level reward design that incorporates material gain, positional advantage, information gain, and strategic structure. Global rewards capture game outcomes; material rewards reflect value-based exchanges; information rewards encourage correctly resolving hidden pieces; and structural rewards promote human-recognizable strategies such as bomb-supported breakthroughs, dense mine-flag defenses, and commander shielding. This reward shaping framework is critical for stabilizing learning in an environment with sparse terminal rewards and long-horizon dependencies.

Summary. Overall, the system demonstrates a unified approach to learning in a complex, partially observable, rule-intensive board game. By combining environment realism, staged DQN agents, belief-driven inference, and domain-informed reward shaping, the project provides an end-to-end pipeline capable of learning meaningful formation patterns, tactical maneuvers, and hidden-information reasoning in *Junqi*.

The core design challenges include constructing a compact state space, handling partially observable information, and shaping rewards to encourage both strategic play and information gain.

2 Game Environment and Rule Implementation

To support reinforcement learning in a partially observable, rule-intensive domain such as *Junqi*, a high-fidelity game environment was implemented to reproduce the complete rule set, board geometry, hidden-information structure, and action constraints of the original game. The environment serves as the foundation for both the placement agent and the reasoning-and-move agent, and provides interfaces for state extraction, legal action generation, battle resolution, and belief update. Its design emphasizes rule correctness, stable state transitions, and compatibility with function approximation.

Board Representation and Spatial Structure. The environment models the official 13×5 *Junqi* board, including camp tiles, fortresses, railway tracks, intersections, and blockable junctions. Railway logic follows the original rules: pieces may traverse multiple consecutive railway cells in a single move, with engineers possessing additional traversal privileges. All grid locations are encoded into a structured state tensor that captures piece occupancy, piece color, visibility status, and traversability information.

Full Rule-Accurate Setup Phase. During the setup phase, each side places its full set of pieces—Commander, Officers, Soldiers, Engineers, Bombs, Mines, and the Flag—within its own camp region. The environment enforces these constraints through:

- legality checks on placement zones,
- preventing occupancy of camp tiles by invalid piece types,
- maintaining per-player occupied-position tables,
- encoding the partially constructed layout as an observation for the placement agent.

This phase concludes only when all pieces have been successfully placed, ensuring formation-level strategies remain consistent across training runs.

Move Phase with Rank-Based Combat and Railway Traversal. Once placement is complete, the environment enters the move-and-attack phase. Legal actions depend on both piece type and board topology. The environment generates all valid actions at each step by considering:

- rank-dependent movement restrictions (e.g., Mines cannot move; Bombs move like regular officers),
- multi-hop railway movement with obstruction checks,
- adjacency-based combat initiation rules,
- piece-specific battle outcomes (e.g., Engineers defusing Mines, Bombs causing mutual destruction).

The battle resolver implements the full hierarchy of *Junqi* combat, handles tie outcomes, and returns both the updated board and visibility updates for the involved pieces.

Hidden Information and Belief Management. A central challenge in *Junqi* is hidden piece ranks: enemy identities remain unknown until either revealed through combat or deduced through logical inference. The environment supports this by:

- masking enemy ranks in the observable state,
- providing a belief tensor for the inference module,
- updating belief maps based on movement histories, combat outcomes, and positional constraints,
- ensuring that both DQN agents operate on partially observable state representations.

This design enables the reasoning agent to learn inference-aware policies that intentionally probe uncertain pieces, avoid risky engagements, and exploit opponent exposure.

Terminal Conditions and Game Flow. Episodes terminate when any of the following occurs:

- the Flag is captured,
- all movable pieces of a player are eliminated,
- no legal moves remain (stalemate),
- forced-draw or deadlock situations occur.

Upon termination, the environment computes global outcome rewards and resets the board for the next self-play episode. This ensures stable episodic learning for both the placement and move agents.

Summary. By combining full rule fidelity, robust hidden-information handling, precisely enumerated legal actions, and clean tensor-based state representation, the environment provides a realistic and stable platform for learning both formation strategies and tactical decision-making in *Junqi*. It forms the core infrastructure upon which the two-stage DQN architecture and belief-based inference system operate.

3 Two-Stage DQN Design

The decision-making process in *Junqi* is inherently multi-phased. The game begins with a fully hidden-information *placement stage*, followed by a long-horizon, partially observable *move-and-attack stage*. These two phases differ sharply in dynamics, action semantics, and reward structure. To accommodate this, the system adopts a two-stage reinforcement learning framework in which two specialized DQN agents—a *Placement Agent* and a *Reasoning & Move Agent*—are trained to solve their respective sub-problems.

3.1 Placement Agent with Self-Reflective Learning

The Placement Agent models formation construction as a sequential decision process in which one piece is placed at a time. The environment exposes an observation tensor encoding the partially completed formation, zone constraints, and remaining unplaced pieces. The agent uses a dueling DQN with noisy exploration layers to evaluate long-horizon formation quality, enabling it to produce robust and deceptive layouts.

3.1.1 State Representation

The placement state is encoded by flattening the board tensor together with a one-hot feature representing the next piece to place. This combines:

- the current partially filled placement board, including color occupancy and camp structure,
- a feature vector indicating which piece is currently selected for placement,
- auxiliary statistics recorded over previous episodes through the reward model (e.g., frequent early defeats, overexposed Commander paths, unstable mine belts).

This design encourages the agent to reason about both immediate placement feasibility and the structural consequences of each decision on the final formation.

3.1.2 Action Space

The Placement Agent operates over a discrete action space indexed by the environment’s `encode_action` function, which maps each valid (piece, cell) pair into a unique ID. Illegal positions (e.g., forbidden camp tiles, occupied cells, or restricted fortress locations) are masked before action selection. Noisy linear layers enable structured exploration in this large and sparse placement space.

3.1.3 Self-Reflective Learning Mechanism

Unlike typical episodic Q-learning, formation quality cannot be evaluated immediately after an action. To address this, the agent employs a *self-reflective learning mechanism*:

- Throughout the placement stage, the agent stores a memory of all intermediate states and actions.
- After the full formation is constructed and the environment computes formation-level rewards (robustness, deception, safety of high-value pieces, structural stability), these rewards are propagated backward across all placement decisions.
- Episode-level reward aggregation penalizes formations that lead to early Commander or Flag exposure, poorly connected mine belts, or chaotic dispersal of high-rank officers.
- Conversely, stable, deceptive, or historically successful formations receive strong positive reinforcement, guiding the agent toward emergent formation patterns.

This reflective mechanism allows the agent to optimize long-horizon formation characteristics that are not directly apparent from single-step actions.

3.2 Reasoning & Move Agent with Rank Inference

Once the initial formation is fixed, the game proceeds to the move-and-attack phase, where long-horizon tactical planning and hidden-information reasoning are essential. The Reasoning & Move Agent is implemented as a belief-aware DQN that integrates both spatial configuration and probabilistic rank inference.

3.2.1 State Representation

The environment provides a structured state embedding consisting of:

- the full spatial board configuration, including own revealed ranks and enemy hidden tokens,
- a battle history buffer encoding all previous engagements and revealed outcomes,
- a dense belief tensor that maintains, for each unrevealed enemy piece, a probability distribution over all legally feasible ranks,
- movement feasibility channels capturing railway intersections, multi-hop lines, blockable points, and adjacency constraints.

This multi-channel tensor provides the agent with both low-level spatial structure and high-level uncertainty modeling required for informed play in hidden-information settings.

3.2.2 Action Space

The action space is again discretized using the environment’s `encode_action` function, mapping each (piece, destination) pair into a unique index. The environment dynamically masks:

- illegal railway multi-hop moves,
- forbidden camp exits,
- movement of immobile pieces (e.g., Mines),
- moves leading into occupied or invalid tiles,
- attacks that violate movement constraints.

This ensures that the agent selects only among legal moves and focuses computation on tactical decision making.

3.2.3 Inference-Driven Q-Learning

A distinguishing feature of the system is its integration of belief-based reasoning directly into Q-value estimation. Before selecting an action, the agent performs:

1. **Belief update:** Using movement histories, combat outcomes, positional constraints, and engineered priors (e.g., Mines never move; Bombs rarely occupy chokepoints), the belief model updates the probability distribution of each enemy piece’s rank.
2. **Q-value conditioning:** The belief tensor is concatenated with the board representation and fed into the dueling Q-network, enabling uncertainty-aware action valuation.
3. **Risk-aware planning:** Moves that preserve high-value pieces, reduce exposure, or strategically probe uncertain enemies are assigned higher Q-values, while reckless or uninformed attacks are discouraged.

This design allows the agent to exhibit sophisticated behaviors such as selective probing, cautious advance under uncertainty, and coordinated breakthroughs supported by bombs or mid-rank officers.

Summary. Together, the Placement Agent and the Reasoning & Move Agent form a unified, multi-stage reinforcement learning pipeline capable of learning both formation-level strategies and fine-grained tactical behavior under hidden information.

4 Reward Shaping and Tactical Bias

Effective learning in *Junqi* requires rewards that reflect not only game outcomes but also the structural, informational, and tactical properties of high-level play. Because terminal rewards are sparse and many crucial decisions occur long before the final outcome, the project introduces a multi-component reward system aligned with the three environments: the setup reward model, the inference reward model, and the gameplay reward model. These reward signals encode both short-term tactical feedback and long-horizon strategic structure.

4.1 Global Outcome Rewards

At the conclusion of each episode, the environment provides a large outcome-based reward:

- a strong positive reward for capturing the opponent’s Flag or eliminating all movable pieces,
- a strong negative reward for losing under the same conditions.

These serve as the primary supervision for long-horizon consistency, anchoring the learning objective to actual win conditions.

4.2 Material, Positional, and Structural Rewards

During the gameplay phase, the reward model evaluates exchanges, territorial changes, and formation integrity. The agent receives:

- **value-scaled material gains** for capturing enemy pieces, with weights reflecting the true *Junqi* hierarchy (e.g., eliminating a Commander or Bomb yields substantially more reward);
- **positional control rewards** for occupying railway intersections, fortress-adjacent squares, supply-line junctions, or central mobility hubs;
- **structural stability bonuses**, such as maintaining a coherent mine belt in front of the Flag, or constructing a multi-layer defensive shell around high-value pieces;
- **loss penalties** for sacrificing high-value officers in low-value trades, exposing the Commander, or breaking the integrity of defensive lines.

Together, these components enforce classical strategic principles such as controlled expansion, stable fortification, and value-based tactical exchanges.

4.3 Information Gain and Inference Rewards

Since *Junqi* is partially observable, the reward model directly encourages information-acquisition behavior.

- The agent receives **positive inference rewards** when its predicted probabilities for an enemy piece’s rank become more accurate after a battle or movement update.
- Rewards scale with the strategic importance of the piece: correctly identifying Bombs, Mines, or the Commander yields larger bonuses.
- The inference reward model additionally provides **entropy-reduction incentives**, encouraging the agent to actively probe uncertain or high-impact enemy pieces.
- The belief model is penalized for **inconsistent assignments** that violate the game’s logical constraints or contradict historical movement patterns.

This encourages the agent to reason about hidden information rather than acting purely reactively.

4.4 Exposure and Information Leakage Penalties

To prevent reckless behavior and unrealistic aggression, the reward system includes explicit penalties for:

- exposing the Commander or Flag through predictable positioning,
- placing Bombs or Mines in easily inferrable patterns (e.g., identical spacing or overly symmetric layouts),
- moving high-value officers into open railway lines without proper escort,
- taking actions that trivially reveal one’s own piece identity.

These penalties enforce information discipline and model the cost of revealing intentions in a hidden-information game.

4.5 Strategic Pattern Guidance

Beyond local tactics, the reward system biases the agent toward recognizable, human-like high-level structures. These include:

- **Commander–Bomb–Officer shielding:** forming a layered defensive shell that protects the Commander from surprise attacks;
- **Bomb-supported offensive breakthroughs:** rewarding formations where a mid-rank officer advances with a Bomb positioned to support high-impact trades;
- **Coherent mine belts and flag enclosures:** penalizing scattered or porous mine structures and rewarding compact ones that restrict enemy access;
- **Railway pressure formation:** encouraging the agent to build multi-unit offensive vectors along railway lines to exert long-range pressure.

These “tactical priors” help the agent avoid local optima and promote holistic strategies that emerge naturally in human expert play.

Summary. Through the combination of global outcome signals, tactical and positional shaping, information-theoretic rewards, and structural strategy priors, the reward design transforms the sparse and highly delayed credit assignment problem of *Junqi* into a tractable RL objective. This reward system is central to enabling the agent to learn meaningful, human-aligned strategies across both the placement and move phases.

5 Results and Insights

Self-play experiments integrating the placement agent, inference module, and belief-aware move agent show that the system learns non-trivial strategies in both formation and combat. Although not yet competitive with expert human play, the agents exhibit meaningful tactical behavior and validate the two-stage RL design.

- **Improved and stable initial formations.** The Placement Agent consistently avoids early-game vulnerabilities, protects high-value pieces, and forms compact defensive structures such as coherent mine belts.
- **Information-aware decision making.** The gameplay agent selectively probes uncertain enemy pieces, avoids high-risk trades when belief entropy is high, and uses railway mobility to gather information before committing to attacks.
- **Emergent tactical patterns.** Behaviors such as bomb-supported breakthroughs, multi-unit railway pressure, and layered defense around the Commander emerge naturally from the reward structure.

While promising, the system still shows limitations: occasional overfitting to specific formations, misinterpretation of ambiguous movement patterns, and inconsistent long-term positional assessment. Nevertheless, for a research-oriented prototype operating in a complex, partially observable game, the results demonstrate solid progress and highlight the effectiveness of combining staged DQN, belief modeling, and structured reward shaping.